

DOCUMENT STRUCTURE (each book must have these fields)

Field	Type
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title	string
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author	string
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genre	string
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price	number
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available	boolean
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pages	number
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year	number
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language	string
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QUESTION : Library Book Management System

Perform the following tasks step by step in MongoDB Compass.

Write each query and its output screenshot in your answer file.

STEP 1 : Create

1. Create a new database named LibraryDB.
2. Inside LibraryDB, create a collection named books.

STEP 2 : Insert

3. Using insertOne(), insert ONE book document with all 8 fields (title, author, genre, price, available,

pages, year, language).

4. Using insertMany(), insert 9 MORE book documents at once.

Make sure your 10 total books include:

- At least 2 different genres
- At least 2 books where available = false
- At least 2 Hindi language books
- At least 3 books with price above 300

STEP 3 : Find / Read

5. Fetch and display ALL documents from the books collection.

6. Using findOne(), fetch only the book whose title matches the first book you inserted in Task 3.

7. Find all books where available is true.

8. Find all books where price is greater than 300.
(Use \$gt operator)

9. Find all books that belong to either "Fiction" or "Fantasy" genre.
(Use \$in operator)

10. Find all books where language is "Hindi" AND price is less than 200.
(Use \$lt operator with two conditions)

STEP 4 : Update

11. Using `updateOne()`, increase the price of any ONE book by changing it to a new value.

(Use `$set` operator)

12. Using `updateOne()`, change the available field of any unavailable book to true.

(Use `$set` operator)

13. Using `updateMany()`, add a new field called rating with value 0 to ALL documents in the collection.

(Use `$set` operator)

14. Using `updateMany()`, increase the price of all books whose genre is "Self-Help" by 50.

(Use `$inc` operator)

STEP 5 : Delete

15. First insert a temporary/dummy book document.

Then delete ONLY that document using `deleteOne()`.

(Show both insert and delete queries)

16. Using `deleteMany()`, delete all books where available is false.

Verify the result by running `find({})` after deletion.

BONUS STEP 6 : Sort, Limit & Projection

17. Display all books sorted by price in ascending order.

(Use `sort()`)

18. Display only the TOP 3 most expensive books.

(Use `sort()` with `limit()`)

19. Display all books showing ONLY title, author and price.

Hide the `_id` field and all other fields.

(Use projection)